

JOSÉ ALFREDO ROSAS CÓRDOVA

👤 October 22, 1999 📍 Guadalajara, Mexico

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EDUCATION

Mechanical Engineering, M. Sc. <i>Leibniz University Hannover</i>	10/2025 — today
Bachelor of Aeronautical Engineering <i>Universidad Autónoma de Nuevo León</i> Graduated summa cum laude	08/2017 — 06/2023
Combined Study and Practice Stays (DAAD-KOSPIE) Mechanical Engineering, B. Sc. <i>Technische Universität Braunschweig</i> GPA: 1.9	09/2021 — 09/2022

LANGUAGES

Spanish	Native speaker
English	TOEFL ITP – 677 Points (C1)
German	TestDaF – TDN 5 (C1)

IT SKILLS

CAD & CAM:	PTC Creo, OnShape, SOLIDWORKS, Fusion 360
Programming:	Python, MATLAB
Simulation & Modelling:	ANSYS Workbench, Fluent, Simulink, Xfoil
Others:	MS Office, Oracle PLM & ERP, L ^A T _E X, Git

WORK EXPERIENCE

Research Assistant <i>Institut für Turbomaschinen und Fluid-Dynamik – Hannover, Germany</i>	04/2026 — today
• Use of specialized software for the extraction of deformation information of a wind turbine with marked blades through the use of digital image correlation techniques. • Post-processing of obtained large-volume datasets using self- and further-developed scripts using Python and MATLAB relying heavily on parallel programming schemes. The processing steps included but were not limited to classification, transformation and mapping of the measured points as well as the generation of diagrams needed for the subsequent validation steps. • Heavy use of Git-based tools to manage software versions and the safe implementation of new functionalities without putting the current production version at risk. • Use of a Blender-based digital twin to validate the correctness of the implemented algorithms before using them to evaluate real-world experimental data.	
Mechanical Engineer - New Product Development & Introduction, Liquid Cooling <i>Vertiv – Monterrey, Mexico</i>	06/2023 — 09/2025
• Development, documentation and sustaining of mechanical components and assemblies encompassing sheet metal welded structures; primary and secondary cooling circuits (i.e. refrigerant and water, copper and stainless steel based piping networks, respectively) for new products tailored towards high density data center cooling applications (e.g. chillers, CDUs, secondary fluid networks). • Experience with design development within a collaborative work environment based on the use of PLM and ERP systems based on the use of Engineering Change Orders and strict revision control. • Execution of both DFM&A as well as DFMEA studies during the design and development process of new products for the identification and correction of areas of opportunity before the parts are committed to production. • Participation in the design validation testing for both complete, production-ready units as well as engineering prototypes for both performance assessments and certification by external entities (e.g. UL, NOM).	

- Collaboration with manufacturing, field teams and facilities for the implementation of design improvements around manufacturing as well as serviceability concerns within both development and production environments.
- Participation in the planning, part procurement, and build process of both engineering, and manufacturing prototypes of brand new products both locally and abroad.

Project Engineer - UAV Development

01/2023 — 04/2023

Spacelab MX – Monterrey, Mexico

- Initial aircraft sizing and preliminary flight performance studies.
- Design of the initial airframe structural and drivetrain components for the prototype phase of the project.
- Integration, ground-, and initial flight testing of the testbed aircraft and its software.

System Testing & Mechanical Design Intern - Engineering Mechatronics

04/2022 — 08/2022

Bosch Rexroth AG – Stuttgart, Germany

- Design and implementation of system tests within a production environment for autonomous mobile robots (UGV) based around previously defined KPIs established around beta-tester customer requirements.
- Evaluation, analysis, and fault detection of the drive performance of the UGVs as well as the communication layer between UGVs and fleet management service based on test data.
- Handling and maintenance of autonomous mobile robots.
- Mechanical design of internal hardware elements and drive mechanisms.

RESEARCH EXPERIENCE

Research Assistant - Aerodynamics Laboratory

07/2020 — 06/2023

Centro de Investigación e Innovación en Ingeniería Aeronáutica – Monterrey, Mexico

- Co-published design case study paper on *Aircraft Engineering and Aerospace Technology* journal. 09/2025
– *Design, manufacture and testing of a low-cost tilt-rotor UAV: a practical development case* (DOI: 10.1108/AEAT-05-2025-0168).
- Co-published potential flow solver validation paper on the *Revista Internacional de Métodos Numéricos para Cálculo y Diseño en Ingeniería* journal. 11/2024
– *Validation of VSPAERO for Basic Wing Simulation* (DOI: 10.23967/j.rimni.2024.10.56492).

EXTRACURRICULAR ACTIVITIES

Club Member

10/2025 — today

Akademischer Fliegergruppe Hannover e.V.

- Participation in the maintenance of the club facilities and aircraft.

Team Member (Technical Report Lead, Chief Engineer)

09/2019 — 06/2022

Axios Aero Design

- Technical report production consisting primarily of a documentation of the engineering design process as well as technical drawings describing the final product to be presented in the yearly SAE competition.
- Aircraft conceptual and preliminary design studies encompassing aerodynamics, structures, electric propulsion, and flight stability and control.
- Project planning and execution based around system engineering methodologies as well as assistance in the detailed design of structural aircraft components such as wing, fuselage, and empennage structures.

AWARDS

SAE Aero Design México 2022

03/2022

2. Place Overall

3. Place Technical Report

MathWorks Minidrone Competition Mexico 2021

08/2021

1. Place

